



Machine Learning

AIML CZG565

M6 : Instance Based Learning

BITS Pilani
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BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Prof.Sugata, Prof.Chetana, , Prof.Monali, Prof.Rajavadhana, Prof.Seetha, Prof.Anita, Prof. Swarna

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction
M2	Machine learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning ✓
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

eager learners

Instance Based Learning

Lazy

Introduction to Machine Learning



Types: Based on how & when the training data is used

- Instance Based Learning: Compare new data points to known data points
- Model Based learning : Detect patterns in the training data and build a predictive model



Lazy Learners : Less time in training
but more time in predicting

Eager Learners : More time in
training but Less time in predicting

Instance Based Learning

Model-based learning techniques

Use the input data

$$\begin{bmatrix} x_{1,0} & x_{1,1} & \dots & x_{1,n} \\ x_{2,0} & x_{2,1} & \dots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m,0} & x_{m,1} & \dots & x_{m,n} \end{bmatrix} \text{ and } \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{bmatrix}$$

X



To learn a set of parameters

$$\begin{bmatrix} \theta_0 & \theta_1 & \dots & \theta_n \end{bmatrix}$$

X



Which yield a **generalized** function

$$f(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$



Capable of predicting values or classes on new input data

$$f(x_i) = 39$$

$$f(x_j) = 1$$

Instance-based learning techniques

Store the input data

$$\begin{bmatrix} x_{1,0} & x_{1,1} & \dots & x_{1,n} \\ x_{2,0} & x_{2,1} & \dots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m,0} & x_{m,1} & \dots & x_{m,n} \end{bmatrix} \text{ and } \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{bmatrix}$$



When asked to predict a new value (a query)

$$y_i = ?$$



Search for similar data points previously stored

$$\begin{bmatrix} x_{4,1} & x_{4,2} & \dots & x_{4,n} \\ x_{9,1} & x_{9,1} & \dots & x_{9,n} \\ x_{15,1} & x_{15,1} & \dots & x_{15,n} \end{bmatrix} \text{ and } \begin{bmatrix} y_4 \\ y_9 \\ y_{15} \end{bmatrix}$$



And use them to generate your prediction

$$y_i = \frac{y_4 + y_9 + y_{15}}{3}$$

Source Credit : <https://www.jeremyjordan.me/k-nearest-neighbors/>

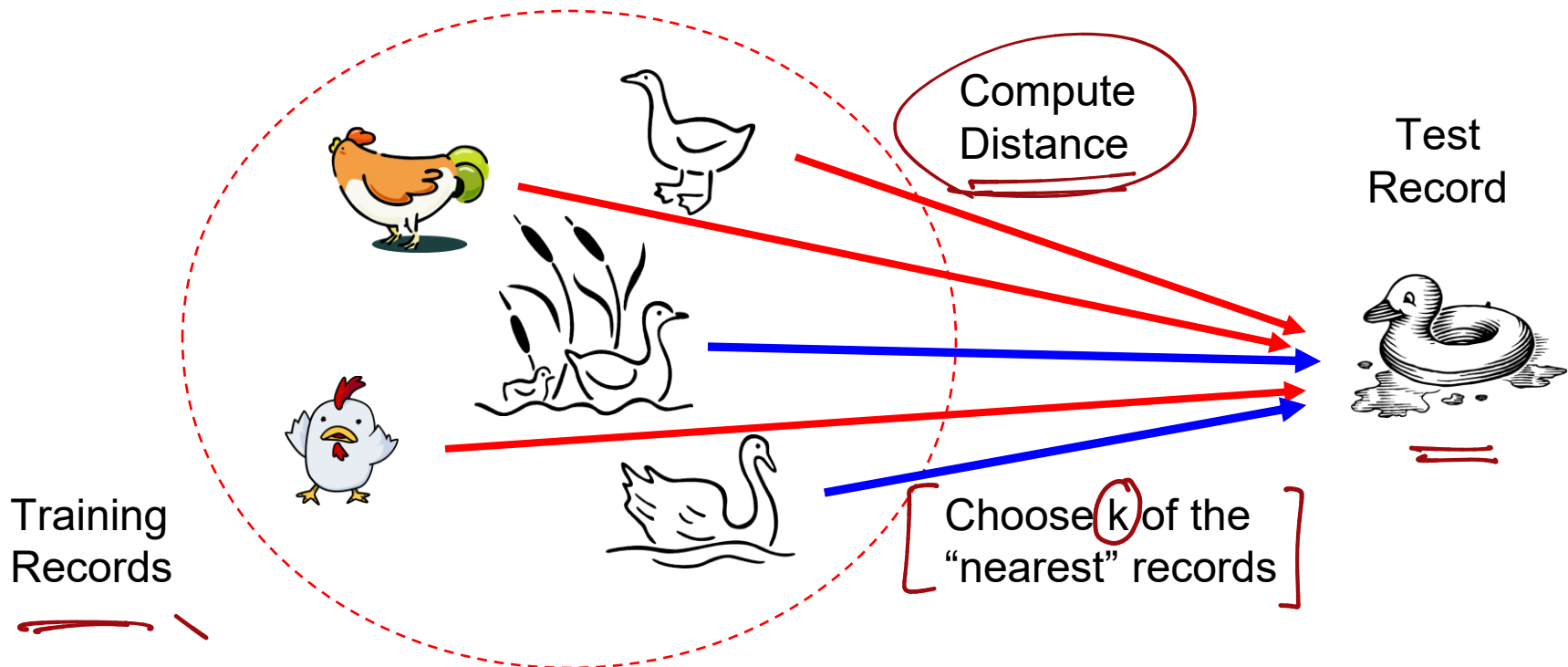
Instance Based Learning



Nearest Neighbor Approach

Basic idea:

- If it walks like a duck, quacks like a duck, then it's probably a duck



Instance Based Learning

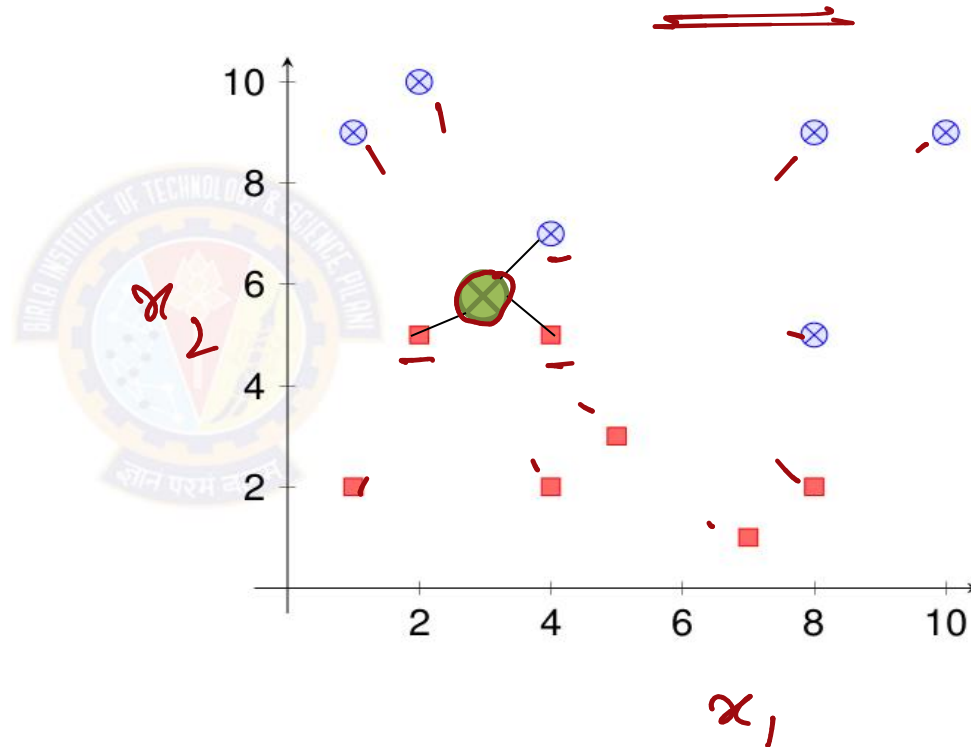


Nearest Neighbor Approach : Notations

K-nearest neighbor:

- Given x_q , take **vote** among its k nearest neighbors (if **discrete-valued target function**)
- Take **mean(or median)** of f values of k nearest neighbors (if **real-valued**) $f^*(x_q) = \sum_{i=1}^k f(x_i)/k$

x_1	x_2	y
1	9	white
10	9	white
4	7	white
4	5	pink
5	3	pink
8	9	white
4	2	pink
2	5	pink
7	1	pink
2	10	white
8	5	white
1	2	pink
8	2	pink



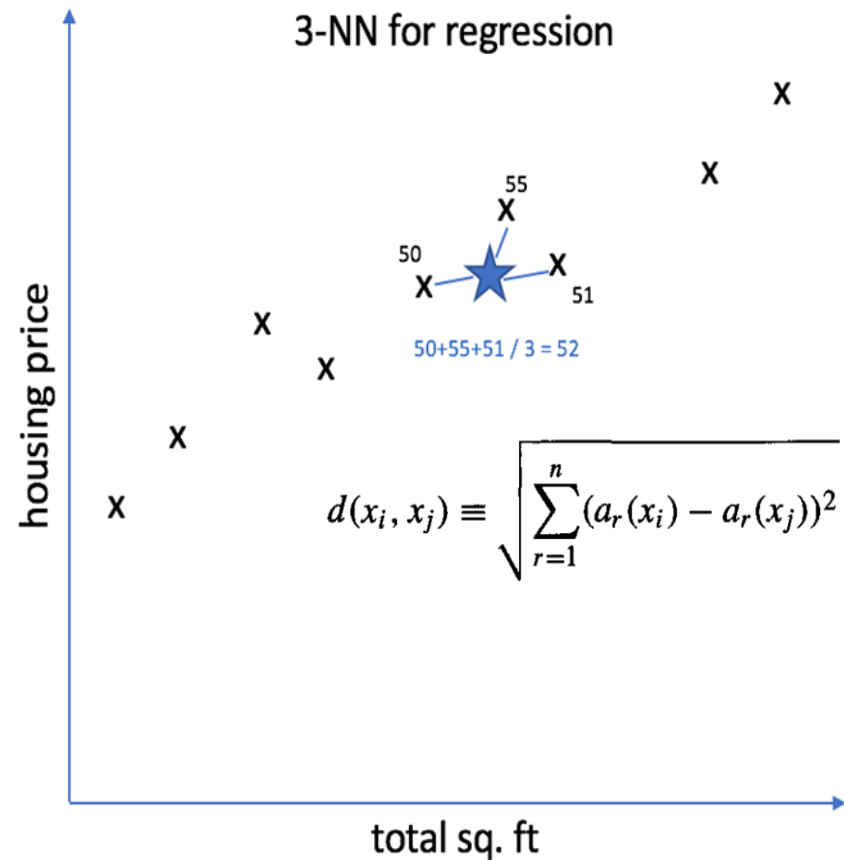
Interpretation : Assume that the Instances represented as points in a Euclidean space

Instance Based Learning



K-Nearest Neighbor Regression: Example

Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
1852	178
...	...
...	...



Source Credit : <https://www.jeremyjordan.me/k-nearest-neighbors/>

Measures of Dis/similarity:

Data Matrix and Distance or dissimilarity

Matrix

Data matrix

- n data points with p dimensions

rows

features

col.

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

Distance / dissimilarity matrix

- n data points, but registers only the distance
- A triangular matrix

p1 p2

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

$d(1,2) = d(2,1)$



Nominal Attributes

Can take 2 or more states, e.g., red, yellow, blue, green (generalization of a binary attribute)

Method 1: Simple matching

- m : # of matches, p : total # of variables $d(i, j) = \frac{p-m}{p}$

i j

Method 2: Use a large number of binary attributes

- creating a new binary attribute for each of the M nominal states

i is red
0/1

i is blue
0/1

i is yellow
0/1

0 to 1

Measures of Distance/similarity: Binary Attributes

- A contingency table for binary data

Object <i>i</i>	Object <i>j</i>		sum
	1	0	
1	<i>q</i>	<i>r</i>	<i>q + r</i>
0	<i>s</i>	<i>t</i>	<i>s + t</i>
sum	<i>q + s</i>	<i>r + t</i>	<i>p</i>

$q = 1 - 1$ $0, 1$
 $0 - 0$
 $0 - 1$
 $1 - 0$

- Distance measure for symmetric binary variables:

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

$\frac{d}{tot}$

when both 0 & 1 are equally valuable

- Distance measure for asymmetric binary variables:

none ←

$$d(i, j) = \frac{r + s}{q + r + s}$$

$\frac{d}{tot}$

- Jaccard coefficient (similarity measure for *asymmetric* binary variables):

$$sim_{Jaccard}(i, j) = \frac{q}{q + r + s}$$

$a_1 \ a_2 \ a_3$
 $i \rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} \ 0 \ 0$
 $j \rightarrow \quad \quad 1 \ 0$

Measures of Dis/similarity: Binary Attributes



Example 1

S

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M ✓	Y -	N	P -	N	N	N
Mary	F ✓	Y -	N	P -	N	P	N
Jim	M	Y	P	N	N	N	N

- Gender is a symmetric attribute
- The remaining attributes are asymmetric binary
- Let the values Y and P be 1, and the value N is 0

9

9 1 2 1 5

$$d(jack, mary) = \frac{\underline{0} + \underline{1}}{\underline{2} + \underline{0} + \underline{1}} = \underline{\underline{0.33}}$$

$$d(jack, jim) = \frac{\underline{1} + \underline{1}}{\underline{1} + \underline{1} + \underline{1}} = 0.67$$

$$d(jim, mary) = \frac{1 + 2}{1 + 1 + 2} = 0.75$$

j

9	1
2	0
1	x

Measures of Dis/similarity



Distances: Minkowski Distance

Minkowski distance: A popular distance measure

L-h norm

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \dots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and h is the order (the distance so defined is also called L - h norm)

Properties

- $d(i, j) > 0$ if $i \neq j$, and $d(i, i) = 0$ (Positive definiteness) ✓
- $d(i, j) = d(j, i)$ (Symmetry) ✓
- $d(i, j) \leq d(i, k) + d(k, j)$ (Triangle Inequality) ✓

A distance that satisfies these properties is a **metric**

Measures of Dis/similarity

Distances : Special Cases of Minkowski

- $h = 1$: **Manhattan** (city block, L_1 norm) **distance**

L-1 norm

✓ — E.g., the Hamming distance: the number of bits that are different between two binary vectors

$$d(i, j) = \underbrace{|x_{i1} - x_{j1}|} + \underbrace{|x_{i2} - x_{j2}|} + \dots + \underbrace{|x_{ip} - x_{jp}|}$$

- $h = 2$: (L_2 norm) **Euclidean** distance

$\sqrt{(x_{i1} - x_{j1})^2 + \dots}$

$$d(i, j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \dots + |x_{ip} - x_{jp}|^2)}$$

- $h \rightarrow \infty$. **“supremum”** ($L_{\max}$ norm, L_{∞} norm) distance.

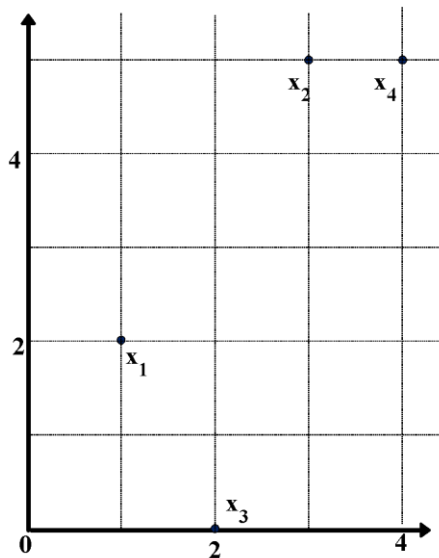
✓ — This is the maximum difference between any component (attribute) of the vectors

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|$$

Example: Minkowski Distance

(Dissimilarity Matrices)

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Manhattan (L_1)

~

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

||

Euclidean (L_2)

~

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

$\sqrt{(\quad)^2 + \quad}$

Supremum

~

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

Standardizing Numeric Data

- Z-score:
 - X: raw score to be standardized, μ : mean of the population, σ : standard deviation
 - the distance between the raw score and the population mean in units of the standard deviation
 - negative when the raw score is below the mean, “+” when above

Where

$$z = \frac{x - \mu}{\sigma}$$

Measures of Dis/similarity : Ordinal Variables

- An ordinal variable can be discrete or continuous
- Order is important, e.g., rank
- Can be treated like interval-scaled

① — replace x_{if} by their rank $r_{if} \in \{1, \dots, M_f\}$

② — map the range of each variable onto $[0, 1]$ by replacing i -th object in the f -th variable by

min - max Norm $\rightarrow$ Number

- compute the dissimilarity using methods for interval-scaled variables

$$\left[z_{if} = \frac{r_{if} - 1}{M_f - 1} \right]$$

Measures of Dis/similarity

Attributes of Mixed Type: Gower distance

- A database may contain all attribute types
 - Nominal, symmetric binary, asymmetric binary, numeric, ordinal

$$d(i, j) = \frac{\sum_{c=1}^n \omega_c \delta_{ij}^{(c)} d_{ij}^{(c)}}{\sum_{c=1}^n \omega_c \delta_{ij}^{(c)}}$$

$d(i, j)$ = dissimilarity between row i and row j ♥

c = the c th column

n = number of columns in the dataset

ω_c = weight of c th column = $\frac{1}{\text{nrows in dataset}}$

$\delta_{ij}^c = \begin{cases} 0 & \text{if column } c \text{ is missing in row } i \text{ or } j \\ 0 & \text{if column } c \text{ is asymmetric binary and both values in row } i \text{ and } j \text{ are } 0 \\ 1 & \text{otherwise} \end{cases}$

$d_{ij}^c(\text{categorical}) = \begin{cases} 0 & \text{if } i \text{ and } j \text{ are equal in column } c \\ 1 & \text{otherwise} \end{cases}$

$d_{ij}^c(\text{continuous/ordinal}) = \frac{|\text{row } i \text{ in column } c - \text{row } j \text{ in column } c|}{\max(\text{column } c) - \min(\text{column } c)}$

<https://healthcare.ai/clustering-non-continuous-variables/>

Example

Based on the information given in the table below, find most similar and most dissimilar persons among them. Apply min-max normalization on income to obtain $[0,1]$ range. Consider profession and mother tongue as nominal. Consider native place as ordinal variable with ranking order of [Village, Small Town, Suburban, Metropolitan]. Give equal weight to each attribute.

Num Nom Nom Ord.

Name	Income	Profession	Mother tongue	Native Place
Ram	70000	Doctor	Bengali	Village
Balram	50000	Data Scientist	Hindi	Small Town
Bharat	60000	Carpenter	Hindi	Suburban
Kishan	80000	Doctor	Bhojpuri	Metropolitan

Solution

After normalizing income and quantifying native place, we get

Name	Income	Profession	Mother tongue	Native Place
Ram	0.67	Doctor	Bengali	1
Balram	0	Data Scientist	Hindi	2
Bharat	0.33	Carpenter	Hindi	3
Kishan	1	Doctor	Bhojpuri	4

$$\frac{2-1}{4-1}$$

$$\begin{aligned}
 d(\text{Ram}, \text{Balram}) &= 0.67 + 1 + 1 + (2-1)/(4-1) = 3 \\
 d(\text{Ram}, \text{Kishan}) &= 0.33 + 0 + 1 + (4-1)/(4-1) = 2.33 \\
 d(\text{Ram}, \text{Bharat}) &= 0.33 + 1 + 1 + (3-1)/(4-1) = 3 \\
 d(\text{Balram}, \text{Bharat}) &= 0.33 + 1 + 0 + (3-2)/(4-1) = 1.67 \\
 d(\text{Balram}, \text{Kishan}) &= 1 + 1 + 1 + (4-2)/(4-1) = 3.67 \\
 d(\text{Bharat}, \text{Kishan}) &= 0.67 + 1 + 1 + (4-3)/(4-1) = 3
 \end{aligned}$$

Most similar – Balram and Bharat; Most dissimilar – Balram and Kishan

Gower's Distance = $d(i,j) / 4$ (with equal weights it becomes simple average)
It will range from 0 to 1.

$$\left[\frac{w_1 d_1 + w_2 d_2 + \dots}{w_1 + w_2 + \dots} \right]$$

≤ 20 ✓
 > 20 ✗

Measures of Dis/similarity

Curse of Dimensionality

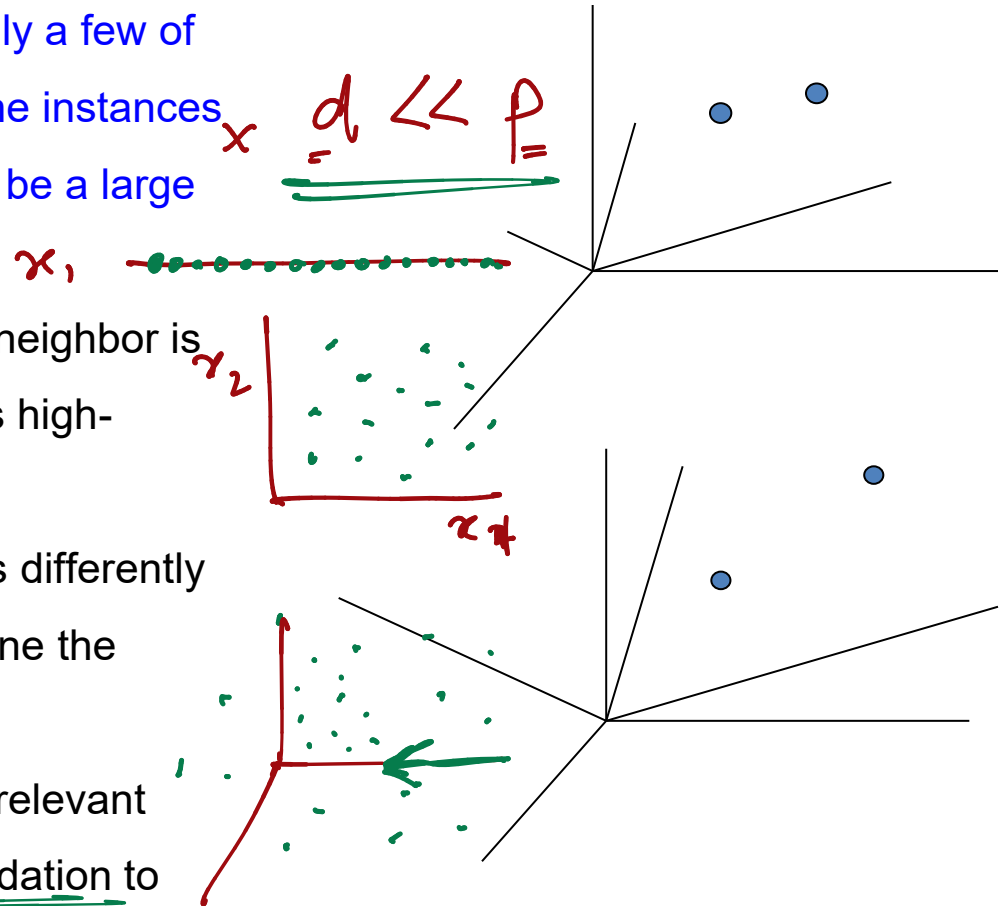
- It uses all attributes to classify instances
- If the target concept depends on only a few of the many available attributes, then the instances that are truly most "similar" may well be a large distance apart.

- **Curse of dimensionality**: nearest neighbor is easily misled when instance space is high-dimensional

✓ **Solution 1:** weigh the attributes differently (use cross-validation to determine the weights)

✓ **Solution 2:** eliminate the least relevant attributes (again, use cross-validation to determine which attributes to eliminate)

Stretching the axes in the Euclidean space, shortening the axes that correspond to less relevant attributes, and lengthening the axes that correspond to more relevant attributes



Instance Based Learning



K-Nearest Neighbor : Algorithm

Approximating a discrete-valued function $f : \mathbb{R}^n \rightarrow V$

Training algorithm:

- For each training example $\langle x, f(x) \rangle$, add the example to the list *training_examples*

Classification algorithm:

- Given a query instance x_q to be classified,
 - Let $x_1 \dots x_k$ denote the k instances from *training_examples* that are nearest to x_q
 - Return

$$\hat{f}(x_q) \leftarrow \underset{v \in V}{\operatorname{argmax}} \sum_{i=1}^k \delta(v, f(x_i))$$

Handwritten notes: A green arrow points to the $\hat{f}(x_q)$ term. A green circle around the k in the summation is labeled 'count'. A red arrow points from the δ function in the summation to the definition below.

where $\delta(a, b) = 1$ if $a = b$ and where $\delta(a, b) = 0$ otherwise.

Approximate a real-valued target function $f : \mathbb{R}^n \rightarrow \mathbb{R}$

$$\hat{f}(x_q) \leftarrow \frac{\sum_{i=1}^k f(x_i)}{k}$$

Handwritten notes: A green bracket is on the left of the equation. A red arrow points from the δ function in the previous block to the $f(x_i)$ term in the numerator.

Handwritten notes: A diagram showing a query point x_q surrounded by its k nearest neighbors $x_1, x_2, \dots, x_k$.

Instance Based Learning



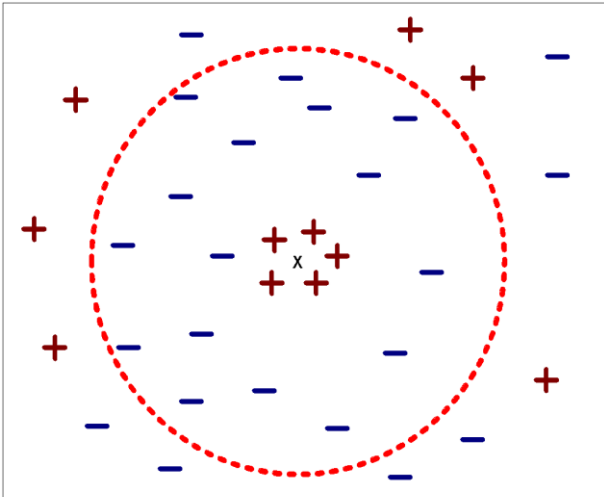
K-Nearest Neighbor : Hyper parameter “K”

Choosing the value of k:

Rule of thumb:

$$K = \sqrt{N}$$

N: number of training points



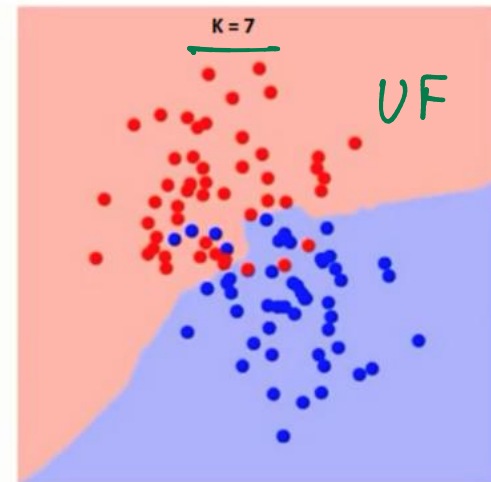
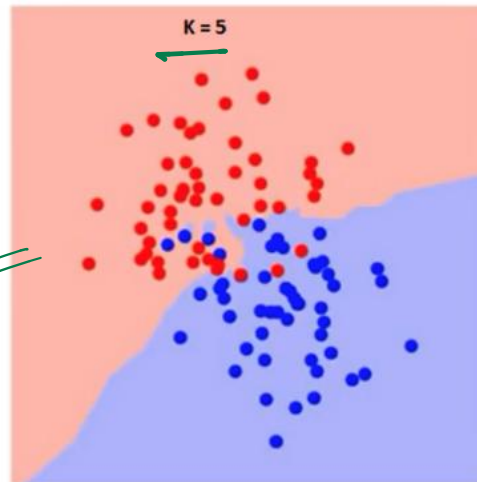
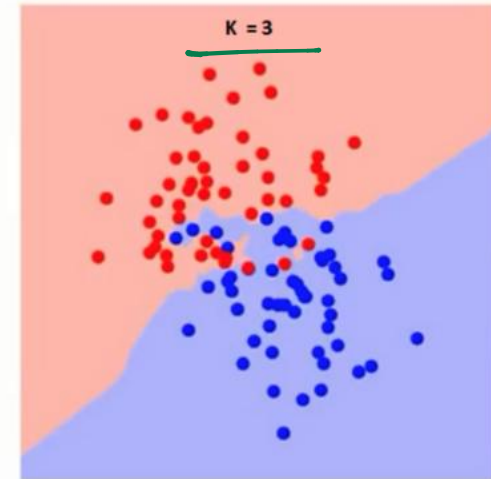
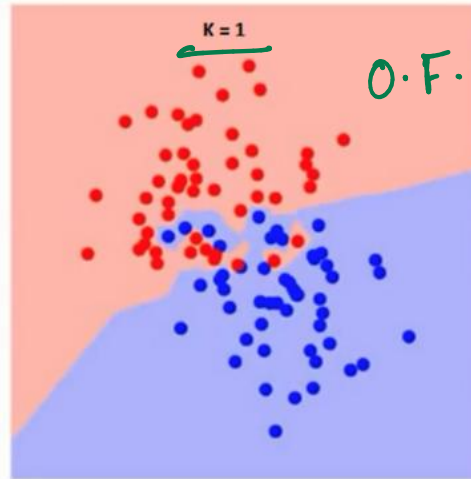
Instance Based Learning



K-Nearest Neighbor : Hyper parameter “K” : Elbow Method

- Compute sum of squares error (SSE) or any other error function for varying values of K (1 to a reasonable X) and plot against K
- In the plot, the elbow (see pic) gives the value of K beyond which the error function plot almost flattens
- As K approaches the total number of instances in the set, error function drops down to '0'

Training data

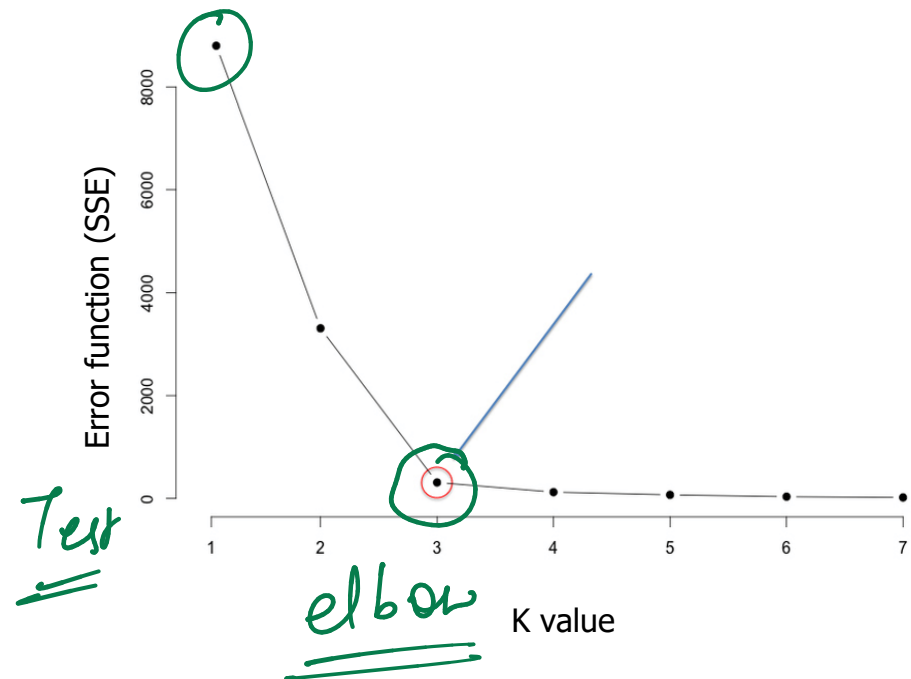


Instance Based Learning



K-Nearest Neighbor : Hyper parameter “K” : Elbow Method

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Instance Based Learning

$K=7$



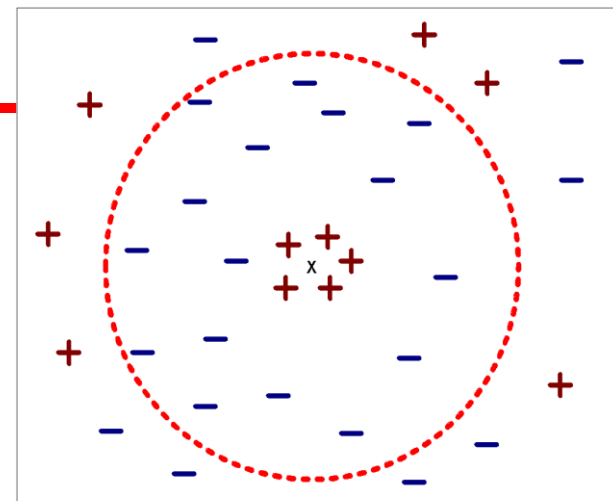
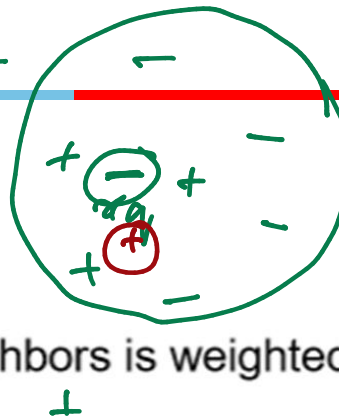
K-Nearest Neighbor : Variation

locally

- Distance Weighted K-NN algorithm

contribution of each of the k nearest neighbors is weighted according to their distance to x_q

- discrete-valued target functions



$$\hat{f}(x_q) \leftarrow \underset{v \in V}{\operatorname{argmax}} \sum_{i=1}^k w_i \delta(v, f(x_i))$$

where $w_i \equiv \frac{1}{d(x_q, x_i)^2}$ and $\hat{f}(x_q) = f(x_i)$ if $x_q = x_i$

- continuous-valued target function:

$$\hat{f}(x_q) \leftarrow \frac{\sum_{i=1}^k w_i f(x_i)}{\sum_{i=1}^k w_i}$$

Kernel Functions

$$w_i = K(d(x_q, x_i))$$

$$K(d(x_q, x_i)) = 1/d(x_q, x_i)^2$$

$$K(d(x_q, x_i)) = 1/(d_0 + d(x_q, x_i))^2$$

$$K(d(x_q, x_i)) = \exp(-(d(x_q, x_i)/\sigma_0)^2)$$

$d(x_q, x_i)$ is the distance between x_q and x_i

$d \rightarrow w$
 $\frac{1}{d^2}$

Problem Type – 1

6 → { 3 +ve
3 -ve

Consider the following training set in 2-dimensional Euclidean space. What is the class of the point (1, 1) if 7NN classifier is considered? If the value of K is reduced whether the class will change? (Consider K=3 and K=5). What should be the final class if the above 3 values of K are considered?

Point	Coordinate	Class	Distance from (1,1)
X1	(-1, 1)	Negative	2 ✓
X2	(0, 1)	Positive ✓	1 ✓
X3	(0, 2)	Negative	1.414
X4	(1, -1)	Negative	2
X5	(1, 0)	Positive ✓	1
X6	(1, 2)	Positive	1
X7	(2, 2)	Negative	1.41
X8	(2, 3)	Positive	2.236

Point	Distance	Class
X2	1	Positive ✓
X5	1	Positive ✓
X6	1	Positive
X7	1.41	Negative
X3	1.414	Negative
X1	2	Negative
X4	2	Negative

$$x_q = (1, 1)$$

$$K = 7$$

$$\sqrt{(1 - (-1))^2 + (1 - 1)^2} = 2$$

7NN
-ve
3NN → +ve
5NN → +ve

Problem Type – 2

$$K = \frac{1}{d(x_q, x_i)^2}$$



Based on the information given in the table below, find the customer type of $x_q = C4$ using K-NN. In given example consider all the instances and use locally weighted K-NN algorithm with kernel $K(d(x_q, x_i)) = 1/d(x_q, x_i)^2$.

Apply min-max normalization on income to obtain $[0,1]$ range. Consider profession and Region as nominal. Consider Locality as ordinal variable with ranking order of [Village, Small Town, Suburban, Metropolitan]. Give equal weight to each attribute.

Customer	Income	Profession	Region	Locality	<u>Category</u>
C0	60000	Doctor	Hindi	Village	L1
C1	70000	Doctor	Bengali	Village	L2
C2	60000	Carpenter	Hindi	Suburban	L2
C3	80000	Doctor	Bhojpuri	Metropolitan	L2
C5	80000	Data Scientist	Hindi	Small Town	L1
C4	50000	Data Scientist	Hindi	Small Town	

x_q

Problem Type – 2

(a) Min–Max normalization of Income

Income range: min = 50000, max = 80000

Income (norm) = (Income – 50000) / 80000

Customer	Income	Normalized ✓
C0	60000	0.333
C1	70000	0.667
C2	60000	0.333
C3	80000	1.000
C5	80000	1.000
C4 (xq)	50000	0.000

Query point C4: 50000, Data Scientist,
Hindi, Small Town

(b) Ordinal encoding for Locality:

Ranking: Village < Small Town < Suburban < Metropolitan

Encoded as: Village = 0, Small Town = 1, Suburban = 2, Metropolitan = 3

To keep equal importance, normalize by max rank (3):

Locality	Normalized
Village	0.000
Small Town	0.333
Suburban	0.667
Metropolitan	1.000

$$\frac{\gamma_i - \gamma_L}{\gamma_{max} - \gamma_{min}}$$

Categorical Distance (Profession, Region)

There are 2 categorical attributes.

$$d_{cat} = \frac{2 - \text{number of matches}}{2}$$

Numerical Distance (Euclidean)

$$d_{num} = \sqrt{(\Delta \text{Income})^2 + (\Delta \text{Locality})^2}$$

Problem Type – 2

Distance computation

C4 and C0:

Numerical part

$$\sqrt{(0.333)^2 + (0.333)^2} = \sqrt{0.222} = 0.471$$

Categorical part

- Profession X
- Region ✓ → 1 match

$$d_{\text{cat}} = \frac{2 - 1}{2} = 0.5$$

$$d(C4, C0) = 0.471 + 0.5 = 0.971$$

C4 and C2:

Numerical

$$\sqrt{(0.333)^2 + (0.334)^2} = \sqrt{0.222} = 0.471$$

Categorical

- Region ✓, Profession X → 1 match

$$d_{\text{cat}} = 0.5$$

$$d(C4, C2) = 0.471 + 0.5 = 0.971$$

Final Distance Table:

Customer	Numerical	Categorical	Total Distance	Category
C0	0.471	0.5	0.971	L1
C1	0.745	1.0	1.745	L2
C2	0.471	0.5	0.971	L2
C3	1.202	1.0	2.202	L2
C5	1.000	0.0	1.000	L1

Majority Vote (Unweighted KNN)

Count class labels:

L1 → C0, C5 → 2 ✓

L2 → C1, C2, C3 → 3

Final Prediction (Normal KNN)

Class of C4=L2

Problem Type – 2

locally wt. dist. $\cdot$ weight $\frac{1}{d^2}$

Final Distance Table (unweighted KNN):

Customer	Numerical	Categorical	Total Distance	Category
C0	0.471	0.5	0.971	L1
C1	0.745	1.0	1.745	L2
C2	0.471	0.5	0.971	L2
C3	1.202	1.0	2.202	L2
C5	1.000	0.0	1.000	L1

Majority Vote (Unweighted KNN)

Count class labels:

L1 → C0, C5 → 2

L2 → C1, C2, C3 → 3

$$\sum w_i \delta_i$$

Final Prediction (Normal KNN)

Class of C4=L2

$$\left[K = \frac{1}{1 + d^2} \right]$$

Final Distance Table (weighted KNN):

Customer	Numerical	Categorical	Total Distance	Weight	Category
C0	0.471	0.5	0.971	1.06	L1
C1	0.745	1.0	1.745	0.33	L2
C2	0.471	0.5	0.971	1.06	L2
C3	1.202	1.0	2.202	0.21	L2
C5	1.000	0.0	1.000	1	L1

Class-wise Weight Sum

Class L1

C0: 1.060

C5: 1.000

Total (L1) = 2.060

Class L2

C1: 0.328

C2: 1.060

C3: 0.206

Total (L2) = 1.594

Final Prediction (Weighted K-NN)

Class of C4=L1



ML 4 marks 10% pre midsem

36 marks 90% post

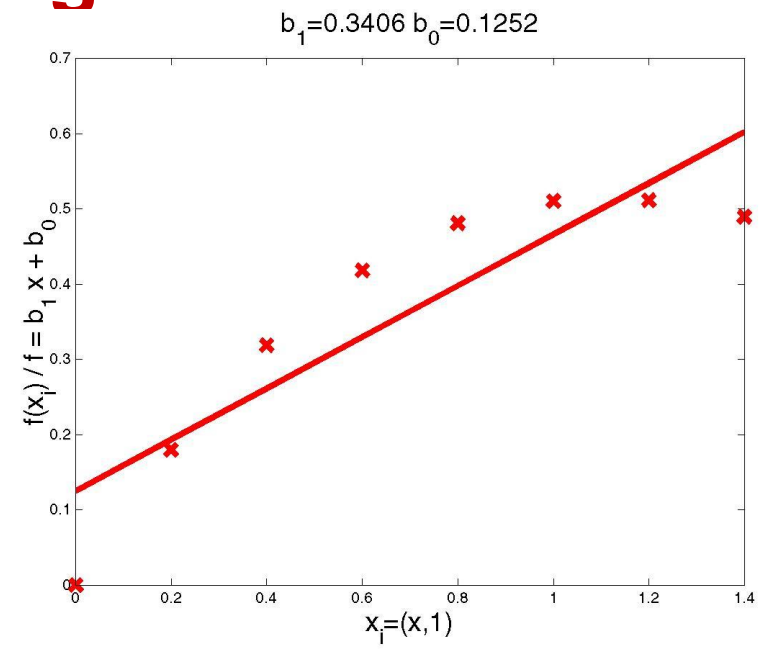
VE - 0
MC - 50%
TC - 50%

PCA correlation analysis

$k=3 \rightarrow \sigma_1$
 $k=5 \rightarrow \sigma_2$
 $k=7 \rightarrow \sigma_3$

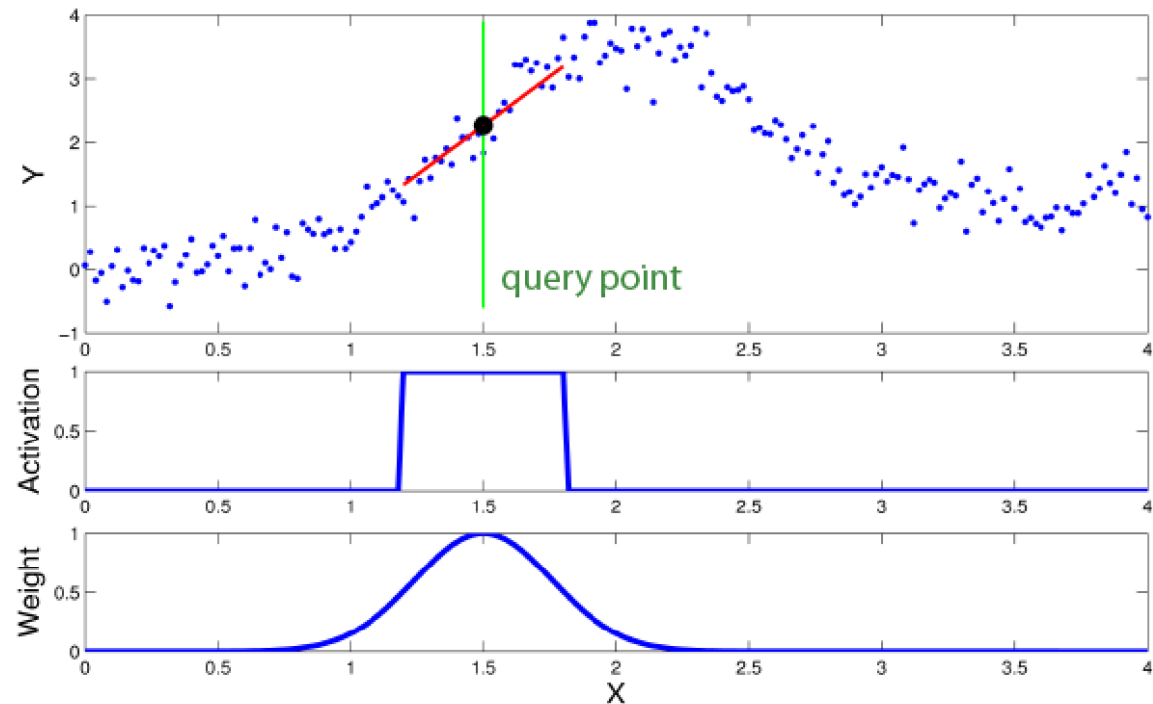
Locally Weighted Regression

[small] training x
 test y $\sim k$



Locally Weighted Regression

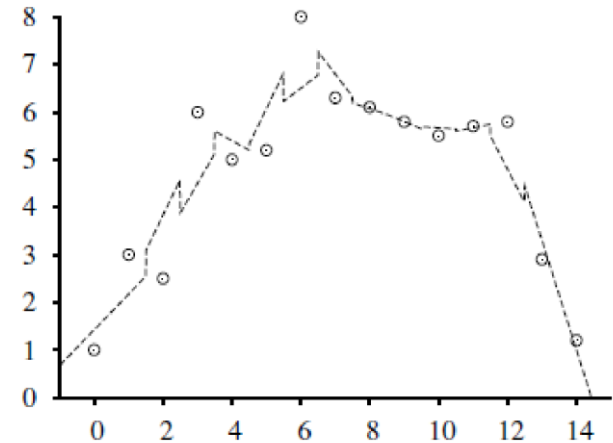
- Locally – Function approximated based on data near query point
- Weighted – Contribution by each training example is weighted by its distance from query point
- Regression- Approximates real-valued target function
- **Residual** is the error in approximating the target function. $\hat{f}(x) - f(x)$
- **Kernel function** is the function of distance that is used to determine the weight of each training example. In other words, the kernel function is the function K such that $w_i = K(d(x_i, x_q))$



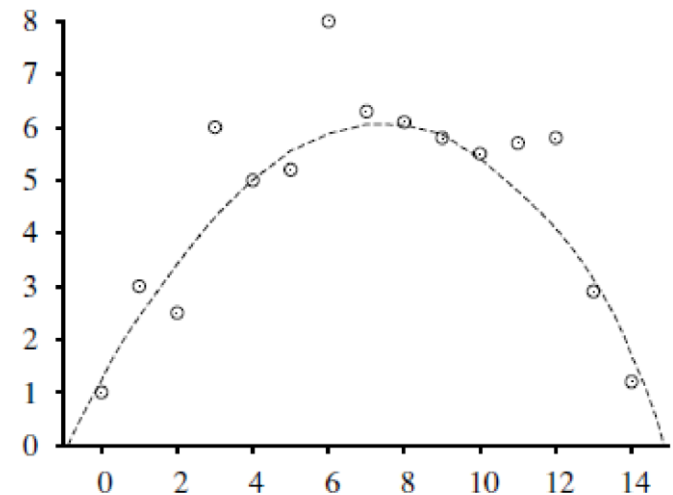
Locally Weighted Regression

- The nearest-neighbor approaches can be thought of as approximating the target function $f(\mathbf{x})$ at the single query point $\mathbf{x} = \mathbf{x}_q$
- Locally weighted regression is a generalization of this approach. It constructs an explicit approximation to f over a local region surrounding $\mathbf{x}_q$
- Approximate the target function in the neighborhood surrounding $\mathbf{x}$, using a linear function, a quadratic function, a multilayer neural network, or some other functional form.

3-nearest-neighbors linear regression



Locally weighted regression with a quadratic kernel



Locally weighted linear regression

- target function is approximated using a **linear function**

$$\hat{f}(x) = w_0 + w_1 a_1(x) + \dots + w_n a_n(x)$$

- methods like **gradient descent** can be used to calculate the coefficients $w_0, w_1, \dots, w_n$ to minimize the error in fitting such linear functions
 - ANNs require a global approximation to the target function
 - here, just a local approximation is needed
- ⇒ the error function has to be redefined

Locally weighted linear regression

possibilities to redefine the error criterion E

1. Minimize the squared error over just the k nearest neighbors

$$E_1(x_q) \equiv \frac{1}{2} \sum_{x \in k \text{ nearest neighbors}} (f(x) - \hat{f}(x))^2$$

2. Minimize the squared error over the entire set D , while weighting the error of each training example by some decreasing function K of its distance from x_q

$$E_2(x_q) \equiv \frac{1}{2} \sum_{x \in D} (f(x) - \hat{f}(x))^2 \cdot K(d(x_q, x))$$

3. Combine 1 and 2

$$E_3(x_q) \equiv \frac{1}{2} \sum_{x \in k \text{ nearest neighbors}} (f(x) - \hat{f}(x))^2 \cdot K(d(x_q, x))$$

Locally weighted linear regression

- For a given query point $\mathbf{x}_q$ we solve the following weighted regression problem using gradient descent training rule:

$$\Delta w_j = \eta \sum_{x \in k \text{ nearest nbrs of } x_q} K(d(x_q, x)) (f(x) - \hat{f}(x)) a_j(x)$$

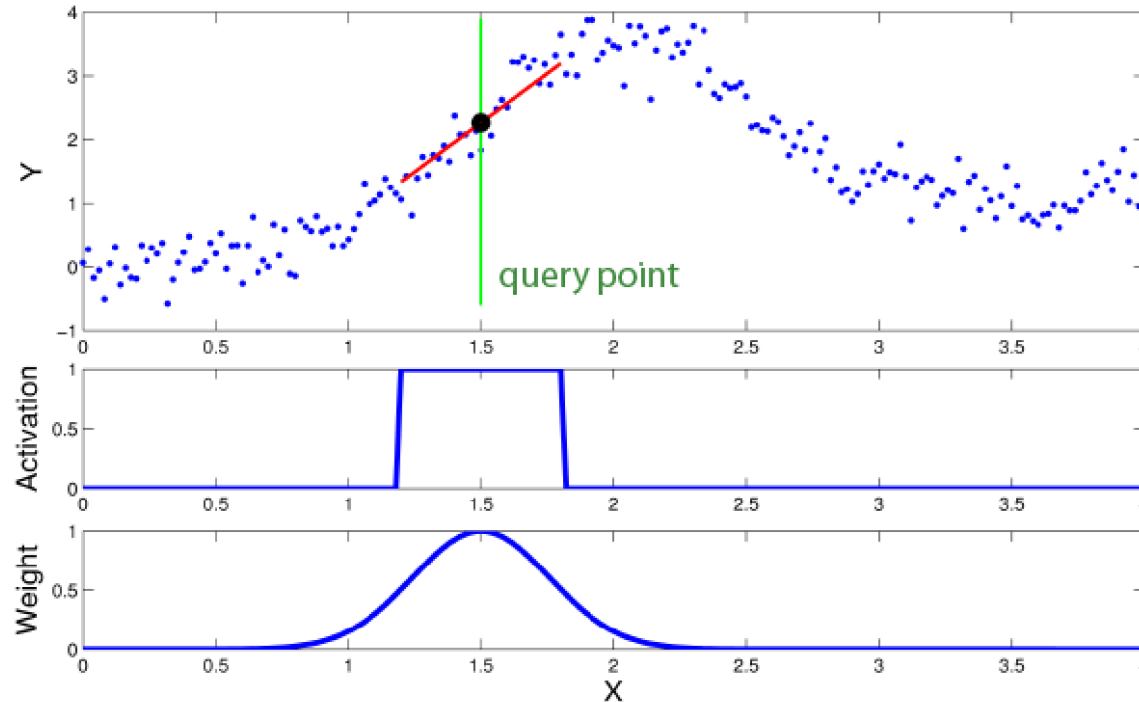
- Note that we need to solve a new regression problem for every query point—that's what it means to be *local*.
- In ordinary linear regression, we solved the regression problem once, globally, and then used the same $\mathbf{h}_w$ for any query point.

$$w_i \leftarrow w_i + \Delta w_i$$

Note: + sign in weight update rule as in Δw_i (actual-prediction) is used.

If we use (prediction – actual) then instead of '+', '-' sign will be used.

Example

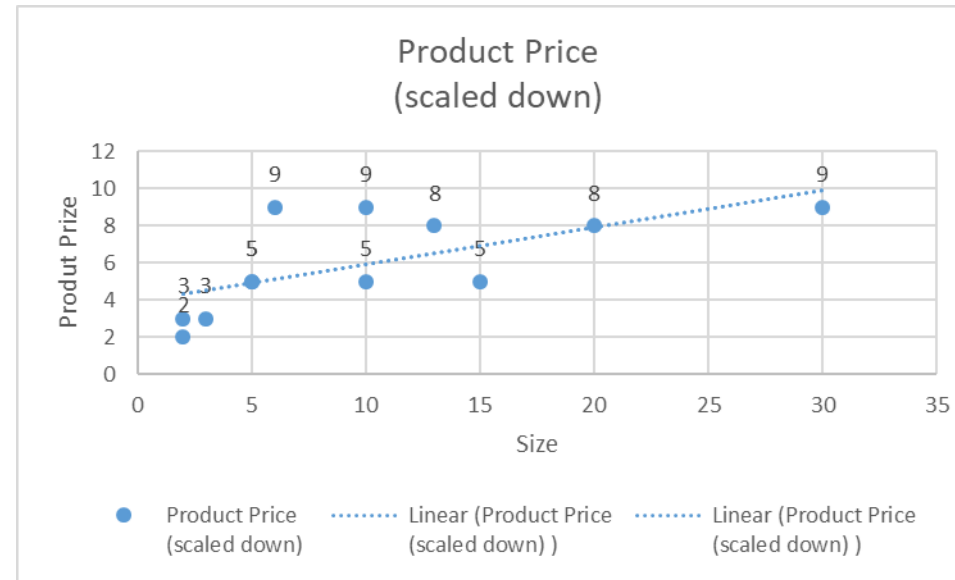


- The upper graphic the set of data points (x,y) (blue dots), query point (green line), local linear model (red line) and prediction (black dot).
- The graphic in the middle shows the activation area of the model.
- The corresponding weighting kernel (receptive field) is shown in the bottom graphic.

Problem Type - 3

Consider the following training set in 2-dimensional Euclidean space. What is the prize of the product with size 7 if 5-NN is considered? Use the kernel of the form $K(d(x_q, x_i)) = \exp(-d(x_q, x_i)^2 / 2b^2)$ where $b=1$. Use the linear regression of the form $y = 3.5 + 0.8x$ and the learning rate of 0.2 to apply the gradient descent and update the weights at the end of first iteration

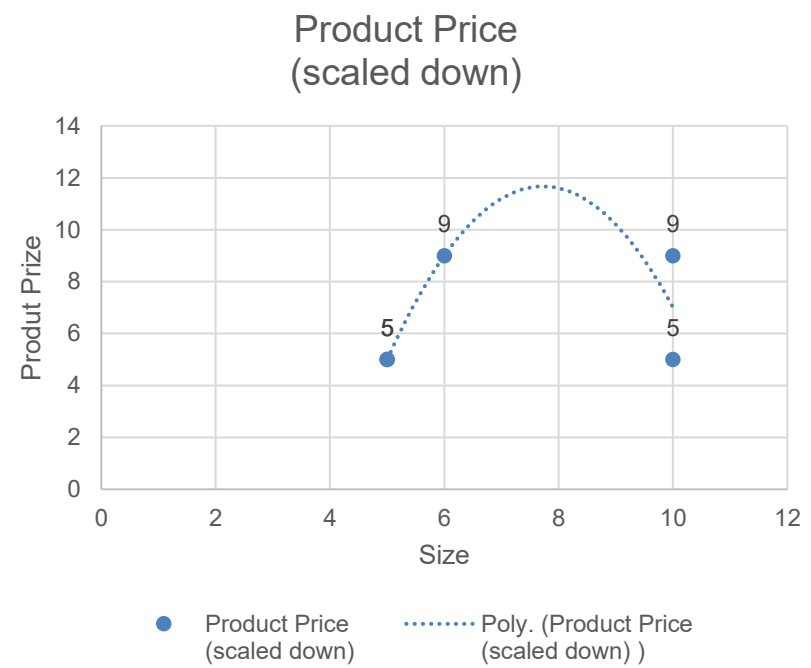
Size	Product Price (scaled down)
10	5
5	5
2	2
3	3
5	5
15	5
30	9
20	8
6	9
2	3
10	9
13	8



Problem Type - 3

Consider the following training set in 2-dimensional Euclidean space. What is the prize of the product with size 7 if 5-NN is considered? Use the kernel of the form $K(d(x_q, x_i)) = \exp(-d(x_q, x_i)^2 / 2b^2)$ where $b=1$. Use the linear regression of the form $y = 3.5 + 0.8 x$ and the learning rate of 0.2 to apply the gradient descent and update the weights at the end of first iteration

Size	Product Price (scaled down)
10	5
5	5
2	2
3	3
5	5
15	5
30	9
20	8
6	9
2	3
10	9
13	8



Problem Type - 3

Use either of below formula :

$$W_{j-new} = W_{j-old} + \alpha * \sum_{i=1}^{K \text{ Neighbors}} K(d(x_q, x_i)) (f(x_i) - \hat{f}(x_i)) a_j(x_i)$$

$$W_{j-new} = W_{j-old} - \alpha * \sum_{i=1}^{K \text{ Neighbors}} K(d(x_q, x_i)) (\hat{f}(x_i) - f(x_i)) a_j(x_i)$$

Size	Product Price (Y)	d(xq, xi)	K(d(x _q ,x _i))	Y'	Y-Y'	(Y-Y') * a	K * (Y-Y') * a	LR * Gradient	W _{old} + delta
10	5	3	0.01	11.5	-6.5	-65	-0.65		
5	5	2	0.14	7.5	-2.5	-12.5	-1.75		
2	2	5	0						
3	3	4	0						
5	5	2	0.14	7.5	-2.5	-12.5	-1.75		
15	5	8	0						
30	9	23	0						
20	8	13	0						
6	9	1	0.61	8.3	0.7	4.2	2.6		
2	3	5	0						
10	9	3	0.01	11.5	-2.5	-25	-0.25		
13	8	6	0						

Sum = -1.8 LR * sum = -0.36 New weight = 0.44

Apply gradient descent where the gradient is weighed by the kernel output.

Answer :

Updated weights after 1st iteration

w0 = 3.43, w1= 0.43

Price Prediction	
6.459515	After 1st iteration of GD
9.1	Before GD

When to Consider Nearest Neighbors

- Suitable for Low dimensional datasets
- Local information can yield highly adaptive behavior
- Lots of training data (distance-weighted KNN)
 - Training is very fast
- Learn complex target functions
 - Do not lose information
- Noisy training data (distance-weighted KNN)
 - by taking the weighted average of the k neighbors nearest to the query point, it can smooth out the impact of isolated noisy training examples.

Nearest-Neighbor Classifiers: Challenges

- The value of k , the number of nearest neighbors to retrieve
- Choice of Distance Metric to compute distance between records
- Expensive, Slow at query time
 - To determine the nearest neighbour of a query point q , must compute the distance to all N training examples
 - + Pre-sort training examples into fast data structures (kd-trees)
 - + Remove redundant data (condensing)
- Storage Requirements
 - Must store all training data **P**
 - + Remove redundant data (condensing)

Additional Exercise for Students

Size	Rating (1-10)	Product Price (scaled down)
10	5	5
5	2	5
2	7	2
3	6	3
5	4	5
15	8	5
30	4	9
20	8	8
6	10	9
2	2	3
10	4	9
13	6	8

1. Consider the following training set in 2-dimensional Euclidean space. What is the prize of the product with size 12 if 7-NN is considered? Use the kernel of the form $K(d(x_q, x_i)) = 1 / d(x_q, x_i)^2$. Use the linear regression of the form $y = 2 + 1.5 x$ and the learning rate of 0.6 to apply the gradient descent and update the weights at the end of first iteration. Use Manhattan Distance for the distance calculation.
2. Try to scale all the features with min-max of [0-5] and redo the modelling under previous question. Observe to Interpret the influence of the scaling
3. Find the MSE of the K-NNs from the results of both above parts separately & state your observations
4. If the Product with features (13,6) and (30,4) are outliers then interpret the effect of the same in the K-NN modelling, before and after removing them with numerical justification.



Lab : Practise :

Take any BITS virtual Lab dataset and experiment on below cases

- **Tune KNN.** Try different values of k to see if you can improve the performance of the algorithm.
- **Use weighted KNN** and check the performance
- **Regression.** Apply it to a regression predictive modelling problem (e.g. predict a numerical value)
- **More Distance Measures.** Implement other distance measures such as Hamming distance, Manhattan distance and Minkowski distance.
- **Data Preparation.** Distance measures are strongly affected by the scale of the input data. Experiment with and without standardization and other data preparation methods in order to improve results.

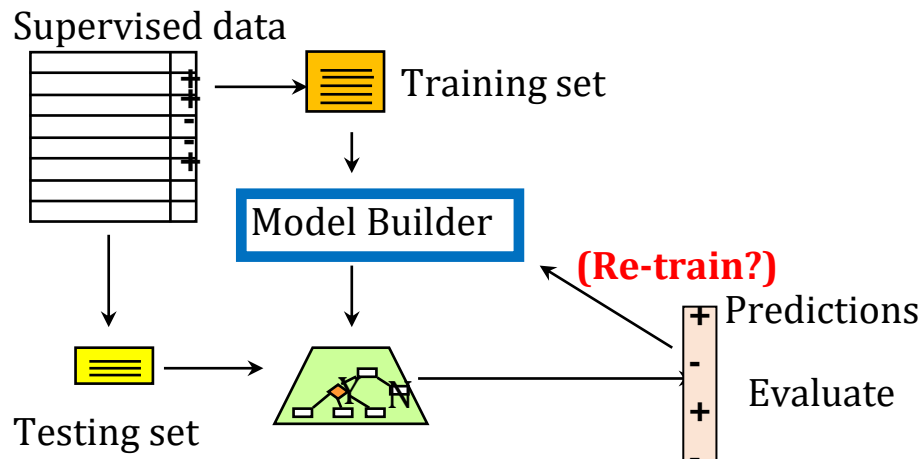
Evaluating Classifier Performance

Model selection and assessment



- Model selection: estimating the performance of different models in order to choose the best one.
- Model assessment: having chosen a final model, estimating its prediction error (generalization error) on new data.
- To estimate generalization error (measure the quality of inductive bias), we need data unseen during training. We split the data as
 - Training set : used to fit the models
 - Validation set : used to estimate prediction error for model selection, tune hyperparameters, feature selection
 - Test (publication) set : is used for assessment of the generalization error of the final chosen model

Hyper parameter tuning using test set



Problem 1: Hesitating between two models (e.g linear model and a polynomial model) ?

Solution 1: train both and compare how well they generalize using the test set: linear model generalizes better but looks like overfitting So, apply some regularization

Problem 2: how to choose the value of regularization hyper parameter?

Solution 2: train 100 different models using 100 different values for this hyper parameter. Choose best hyper parameter value with the lowest generalization error (5%). Great!!!: launch into production

Problem 3: Error now is increased to 5% errors. Why?

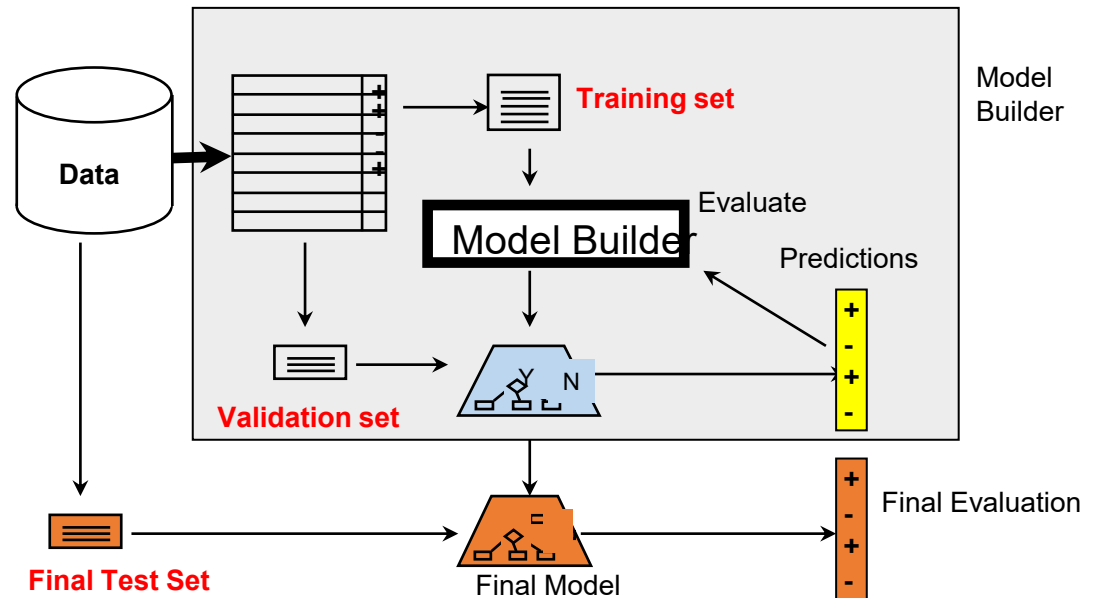
Answer 3: Generalization error was measured multiple times on the test set, So model and hyper parameters are adapted to produce the best model for that particular set, unlikely to perform as well on new data.

Learning : The test data can't be used for parameter tuning!

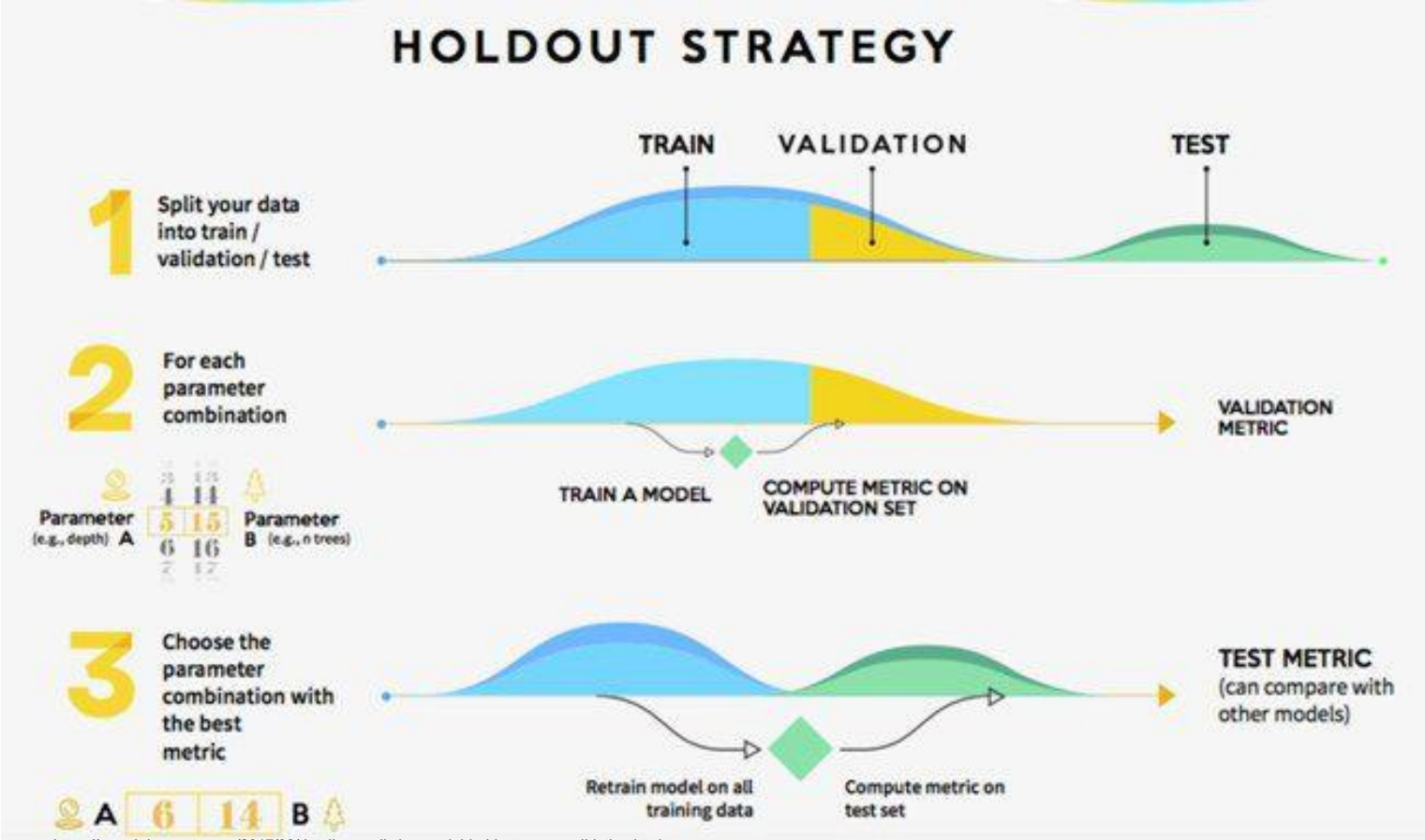
Train, Validation, Test split

Holdout validation (holdout or development, or dev set)

1. Train multiple models with various hyper parameters on the reduced training set (i.e., the full training set minus the validation set)
2. Select the model that performs best on the validation set
3. After this holdout validation process, you train the best model on the full training set (including the validation set), to get the final model.
4. Lastly, evaluate this final model on the test set to get an estimate of the generalization error



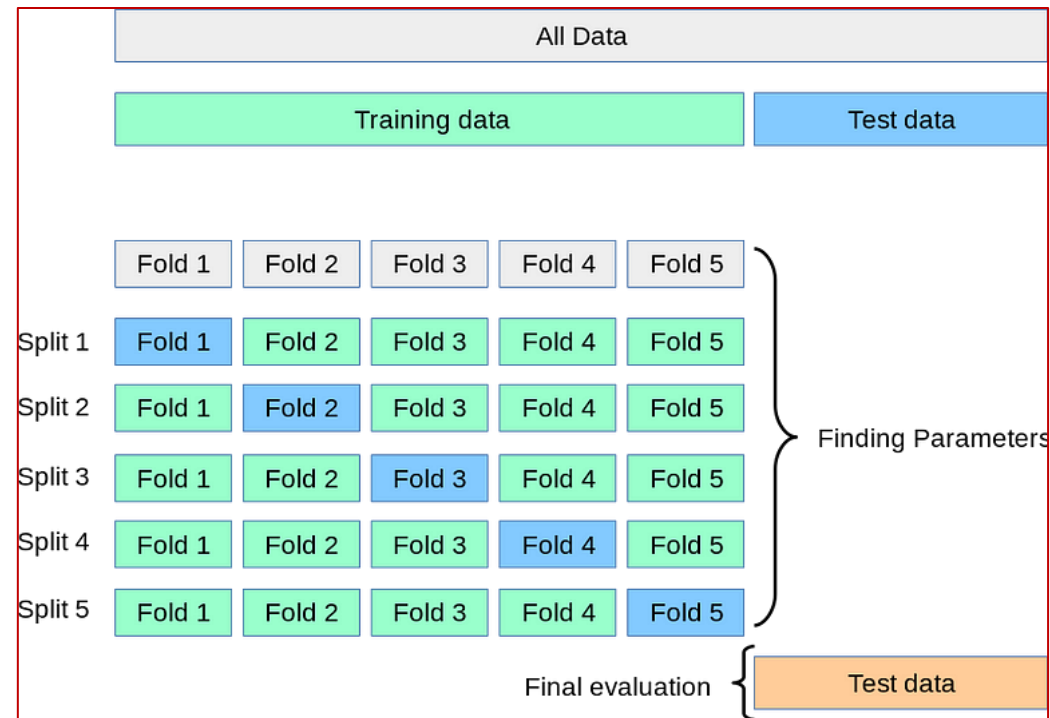
Train-Test-Validation



<https://www.kdnuggets.com/2017/08/dataiku-predictive-model-holdout-cross-validation.html>

Making most out of the data

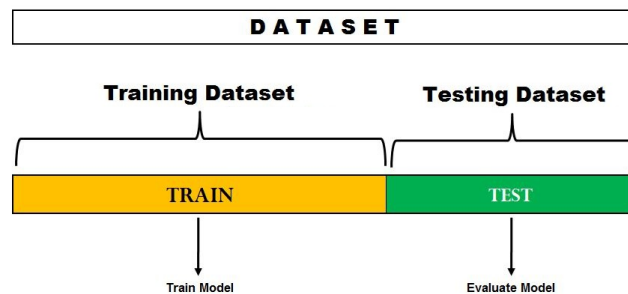
- We don't want to touch the test set until we are ready to launch a model we are confident about
- If the validation set is too small, then model evaluations will be imprecise
 - Larger the test data the more accurate the error estimate
- If the validation set is too large, then the remaining training set will be much smaller than the full training set.
 - Larger the training data the better the classifier



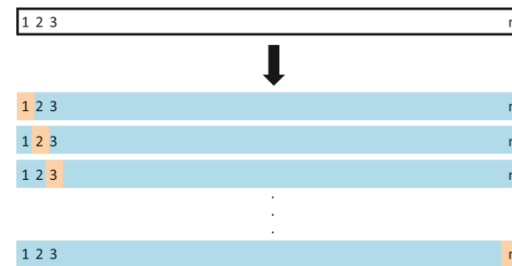
- **Solution: cross validation**

Cross-Validation techniques

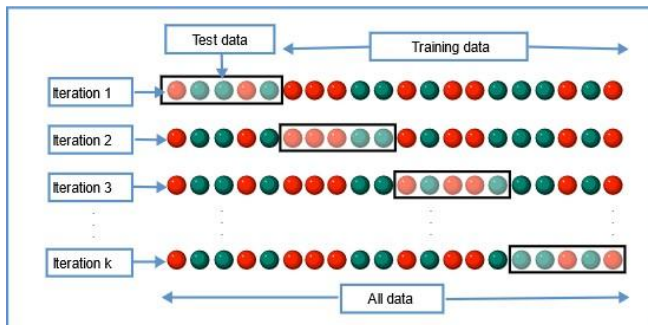
Hold Out method



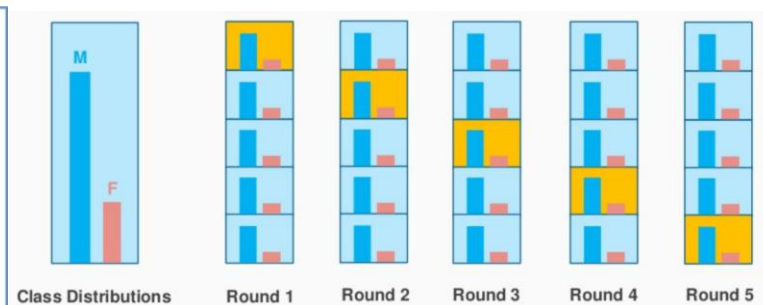
Leave One Out Cross-Validation



K-Fold Cross-Validation



Stratified K-Fold Cross-Validation

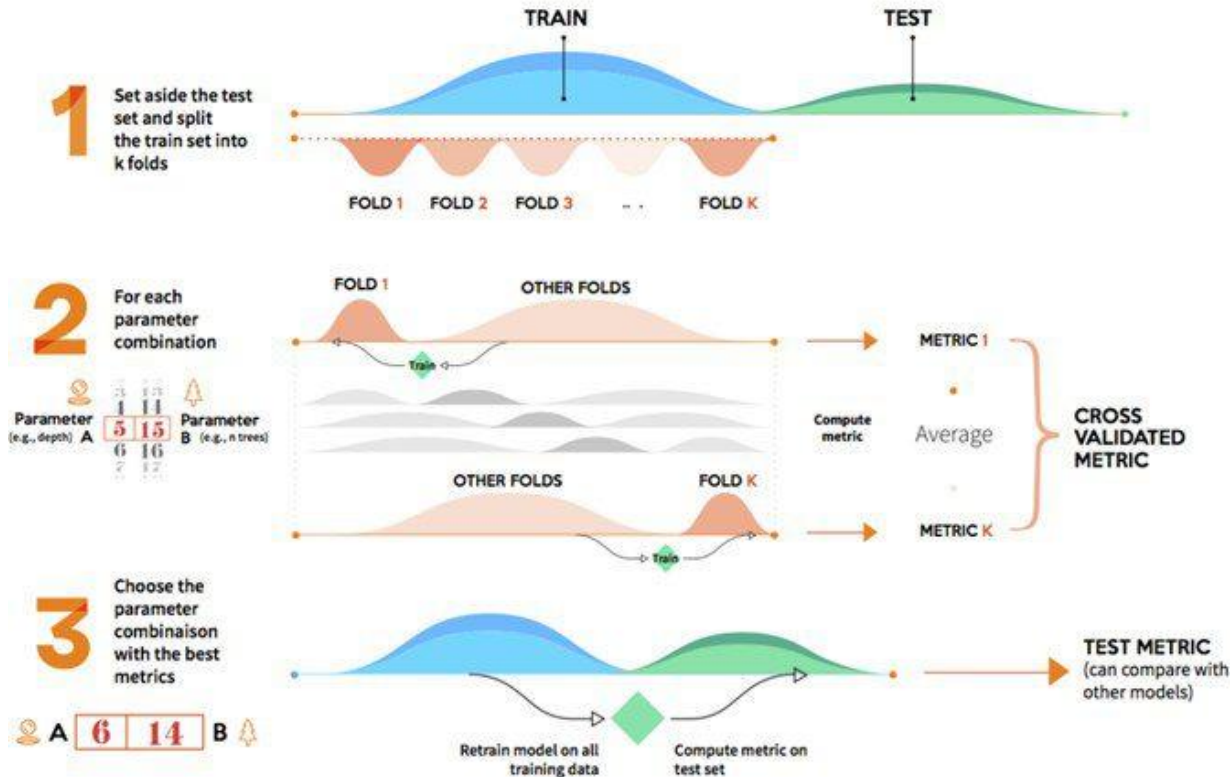


<https://www.analyticsvidhya.com/blog/2021/05/4-ways-to-evaluate-your-machine-learning-model-cross-validation-techniques-with-python-code/>

Cross-Validation for hyper parameter tuning



K-FOLD STRATEGY



<https://www.kdnuggets.com/2017/08/dataiku-predictive-model-holdout-cross-validation.html>

Estimating generalization error : cross validation set Vs test set

- Training set performance is not a good indicator of generalization error
 - A more complex model overfits, a less complex one underfits
 - Which model do I select?

- Test set
 - Typically 80%, 20%
 - Wastes valuable labeled data

- Cross validation
 - In Sklearn : `cross_val_score()` (default is accuracy)
 - `cross_val_score` will automatically split data into train and test
 - fit the model on train data
 - score on test data
 - For 10 folds, 10 scores will be returned in scores variable. Then take average of that

Split 1	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Metric 1
Split 2	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Metric 2
Split 3	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Metric 3
Split 4	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Metric 4
Split 5	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Metric 5

Training data Test data

```
model=DecisionTreeClassifier()  
scores = cross_val_score(model, X, y, cv=10)  
avg_score = np.mean(scores)  
print(avg_score)
```

Thank you !

Required Reading for completed session :

T1 - Chapter # 8 (Tom M. Mitchell, Machine Learning)

Next Session Plan :

Module 7 : Support vector machines will be covered in CS 15 or 16

Important Note for SVM : Below Math Pre-requisite (Yet to be discussed in MFML parallel course) :
Since the math basics will be covered in detail in math course, ML course is intended to discuss the application & interpretation of the model

Lagrange Multipliers

KKT – Conditions

Primal – Dual for SVM

Kernel for Non-Linear SVM